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AI Use by the Judiciary: What SC's Proposed Regulations Allow, What Is Barred

AI use by the judiciary: What SC's proposed regulations allow, what is barred



EXPERT EXPLAINS

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THE SUPREME Court (SC) last month released the draft of Regulations for Use of Artificial Intelligence (AI) in Courts, 2026, aimed at developing a governance framework for use of AI in the judiciary, prescribing general principles regarding AI's use, and establishing an institutional framework for the same. It has sought comments and suggestions from stakeholders and the public by July 15.

The Draft Regulations may be seen as an effort to regulate the use of AI in court processes to optimise court administration.

Is this binding?

Not automatically, and not uniformly. The Draft Regulations will come into force for the SC on any date notified by the Chief Justice of India. For High Courts and courts and tribunals under their jurisdiction, they will come into force separately on the dates notified by that High Court's Chief Justice. Different provisions can be brought into force on different dates, meaning that AI adoption can be further phased to suit the courts' requirements and circumstances.

What role for AI has the SC carved out?

The Draft Regulations propose to require courts to "actively seek opportunities" to deploy AI systems and tools that "demonstrably" improve access to justice, reduce delays, or enhance administrative efficiency.

AI use is explicitly permitted for a range

of administrative and assistive functions, including case management, transcription, translation, legal research, document summarisation, accessibility, and court administration. All of these will require approval in writing by the Apex Body of the SC, or AI Committee of the high court/tribunal, as the case may be, and supervision and verification of officers nominated by the court.

Can AI decide a case?

No. The Draft Regulations are categorical that no judicial outcome shall be reached through algorithmic decision-making alone, or solely on the basis of AI-generated information. Human judicial authority is determinative in all adjudicative decisions. Therefore, the use of AI in the decision-making process related to a case, if any, will only be advisory, and subject to in-

dependent human (judicial) evaluation.

What is off-limits?

Several uses are prohibited in "absolute and non-derogable" terms, meaning no authority can later permit them under the Regulations. These include: 'risk scoring' to assess flight risk; predicting recidivism; evaluating bail eligibility; determining the credibility of witnesses; using AI to predict, profile, or infer the future conduct or behaviour of parties, accused persons, witnesses, or legal representatives more generally; submitting AI-generated output as independent evidence without full disclosure of its AI-generated character; and using

blackbox (unexplainable) AI systems in matters affecting personal liberty.

Will litigants know if AI was used for



A weekly column decoding AI's new frontiers, how they shape the world and our place in it

their case?

Yes, the Draft Regulations propose that if a court uses an AI tool to "materially assist" it in any aspect of case management, document analysis, or judicial administration, it will be required to inform the parties (litigants and their counsel) in a timely and accessible manner. The standard is material assistance; therefore, litigants will not be informed of all instances where AI has been used in their case, but only when AI has provided material assistance to the judge.

Who will regulate?

The institutional architecture will be led at the highest level by an "Apex Body" at the SC, composed of sitting SC and High Court judges, an official of the Ministry of Electronics and Information (MeitY), and experts in finance and cybersecurity. This body will set out the minimum mandatory standards approving of the use of AI systems and issuing implementation guidelines. It will work through five specialised

committees; the SC and each High Court will constitute their own AI Committees, backed by an AI Secretariat. A separate research body, the Centre of Research and Excellence on Artificial Intelligence (CoRE-AI), will evaluate tools and track developments to support the Apex Body.

Will private companies be involved?

Yes, but only with written approval from the relevant court authority, and subject to a mandatory list of contract terms. Vendor agreements must cover ownership of and access rights to court data and AI outputs, a bar on using sensitive judicial data, and an explicit prohibition on retaining or fine-tuning models using court data without the AI Committee's written approval.

The Draft Regulations also state that vendors cannot claim exclusive IP over tools built substantially on judicial data or public resources.

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AI Use by the Judiciary: What SC's Proposed Regulations Allow, What Is Barred

- **The Draft Regulations**

- The Supreme Court released the draft Regulations for Use of Artificial Intelligence (AI) in Courts, 2026, aiming to develop a governance framework for AI in the judiciary; public comments sought till July 15

- **Not Automatically Binding**

- The Regulations come into force for the SC on a date notified by the CJI; High Courts and subordinate courts adopt them separately on dates set by each High Court's Chief Justice

- **Permitted Uses**

- AI is explicitly allowed for administrative and assistive functions : case management, transcription, translation, legal research, document summarisation, accessibility and court administration

- **What Is Off-Limits**

- AI cannot decide a case; prohibited uses include risk scoring for flight risk, predicting recidivism, evaluating bail eligibility, judging witness credibility, and using **black box AI** in matters affecting personal liberty

- **Institutional Architecture**

- Regulation led by an "Apex Body" at the SC (sitting judges, a MeitY official, finance/cybersecurity experts); each court gets its own AI Committee, backed by a new Centre of Research and Excellence on AI (CoRE-AI)

AI Use by the Judiciary: What SC's Proposed Regulations Allow, What Is Barred

Terms related to AI

Weak AI/ Narrow AI: It is a type of AI that is limited to a specific or narrow area. Weak AI simulates human cognition.

Strong AI: These are systems that carry on tasks considered to be human-like. These tend to be more complex and complicated systems.

Augmented Intelligence systems are designed to work alongside humans to improve their ability to perform tasks.

Theory-of-mind AI: They are fully adaptive and have an extensive ability to learn and retain past experiences. These types of AI include advanced chat-bots that could pass the Turing Test, fooling a person into believing the AI was a human being.

A **Turing test** is a method of inquiry in AI for determining whether or not a computer is capable of thinking like a human being.

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Black Box AI : AI whose decision-making process is difficult or impossible for humans to understand.

Explainable AI (XAI) : AI that provides reasons or explanations for its decisions.

Agentic AI : AI that can plan, make decisions, and perform tasks independently to achieve a goal.

Generative AI : AI that creates new content such as text, images, music, videos, or code.

AI Hallucination : When an AI generates false, fabricated, or misleading information while presenting it as if it were true.

Data Mining : The process of extracting meaningful patterns or knowledge from large datasets.

Bishnoi Indictment: The Long Road to Extradition

Bishnoi indictment: The long road to extradition



DEEPTIMAN TIWARY

DAYS AFTER indicting gangster Lawrence Bishnoi in a series of crimes, including the killing of Canada-based Khalistan separatist Hardeep Singh Nijjar, the US Department of Justice (DoJ) has confirmed to *The Indian Express* that it will seek his extradition.

Last week, the US DoJ had charged 37 people, also including Punjab Police SHO Gurinderjit Singh Nagra, in the indictment as part of its 'Operation Hardball' — a coordinated action by law enforcement from the US, Canada and Europe which led to the arrest of 24 people linked to three Indian crime syndicates with transnational networks. The charges included racketeering conspiracy, murder and extortion.

When the US does formally submit its extradition request, what legal process will be triggered? Do the charges qualify for extradition under the India-US treaty? Can India refuse to hand over Bishnoi — lodged in a jail in Gujarat — or at least delay his surrender? The answers lie in the India-US Extradition Treaty, India's Extradition Act, 1962, and a legal process in which Indian courts and the Union government both play distinct roles.

What governs extradition between India and the US?

Extradition between India and the US is governed by the bilateral Extradition Treaty signed in 1997 and India's Extradition Act, 1962. At the heart of the treaty is the principle of dual criminality — an offence is extraditable if it is punishable with imprisonment of more than one year in both countries.

The treaty bars extradition for political offences, although offences such as murder, hostage-taking, terrorism and drug trafficking are specifically excluded from that exception. Nor can extradition be refused merely because the person sought is an Indian citizen. Another safeguard is the rule of speciality, under which an extradited person can ordinarily be tried only for the offences for which extradition was granted.



Lawrence Bishnoi. FILE

EXTRADITIONS BETWEEN INDIA AND THE U.S. ARE INFREQUENT

11 Fugitives extradited by the US to India between 2002 and 2018

60 Pending Indian requests with US authorities in this time

Key extradition requests to US

REFUSED



David Coleman Headley
26/11 accused
Why: His plea agreement barred it

CLEARED



Warren Anderson
Union Carbide chief
Why: US cited deficiencies in the evidence



Tahawwur Rana
26/11 accused
But: Came after a lengthy legal process

Dual criminality

At the heart of the India-US extradition treaty, signed in 1997, is the principle of 'dual criminality'.

This means an offence is extraditable if it is punishable with jail of more than one year in both countries.

What happens if Washington sends an extradition request?

Typically, an extradition request would be prepared by the US DoJ and transmitted by the State Department to India's Ministry of External Affairs (MEA). The MEA, in consultation with the Ministry of Home Affairs and agencies such as the CBI, would examine whether the request conforms to the treaty and the Extradition Act.

If the Centre decides the request merits consideration, it is placed before an Indian court, which examines whether the treaty requirements are satisfied and whether the material produced by the US would justify committing had the offence been committed in India.

If satisfied, the court certifies the accused as extraditable and forwards its findings to the Central government. The final decision on surrender nevertheless rests with the Union government, which may impose conditions or seek diplomatic assurances before approving extradition. Any order can then be challenged before the High Court and the Supreme Court, making extradition proceedings inherently lengthy.

Rana's case illustrates this well. India's request, made in 2019, was followed by

The US is set to seek the extradition of gangster Lawrence Bishnoi. But a formal request is only the first step in a lengthy legal and diplomatic process

proceedings before a US magistrate judge, habeas corpus proceedings before a district court, appeals before the Ninth Circuit and petitions before the US Supreme Court before his extradition was finally approved. Even a priority terrorism case took years to conclude.

Can India refuse or delay extradition?

Even if the treaty requirements are satisfied, extradition is not automatic. Meeting the treaty's conditions merely enables the process; it does not compel India to immediately surrender the accused.

For Bishnoi in particular, the most significant legal consideration is that he is already in judicial custody and faces numerous criminal prosecutions across India.

Article 14 of the treaty says the requested state "may" temporarily surrender the accused for trial before requiring his return. But read with Section 31 of the Extradition Act, which permits India to postpone surrender while domestic proceedings are pending, it gives New Delhi considerable discretion over the timing of any extradition.

This means India could legally argue that Bishnoi must first face trial — and, if convicted, serve any sentence — in the numerous murder, extortion and organised crime cases pending against him before any extradition takes place.

The Punjab Police SHO Nagra stands on a different footing as he is not facing comparable prosecutions in India. Even so, Indian authorities could initiate domestic proceedings if the conduct alleged in the US indictment also discloses offences under Indian law.

Section 34 of the Extradition Act deems an extradition offence committed abroad to have been committed in India, while Section 34A empowers the Centre to prosecute a fugitive in India if it decides he cannot be surrendered.

The Vikash Yadav case may provide a clue. After US prosecutors indicted the former RAW officer over the alleged plot to assassinate Khalistani separatist Gurpatwant Singh Pannun, Delhi Police registered an FIR and arrested him. If Washington eventually seeks Bishnoi's extradition, India will have to decide whether domestic prosecution should take precedence over extradition — a question that could equally arise in Nagra's case.

Bishnoi Indictment: The Long Road to Extradition

- **The Development**

- Days after indicting gangster Lawrence Bishnoi in a series of crimes including the killing of Canada-based separatist Hardeep Singh Nijjar, the US Department of Justice confirmed it will seek his extradition

- **Legal Framework**

- Extradition between India and the US is governed by the bilateral Extradition Treaty (1997) and India's Extradition Act, 1962; central to the treaty is "**dual criminality**": the offence must be punishable by over one year's imprisonment in both countries

- **Historical Pattern**

- Extraditions between India and the US are infrequent: only 11 fugitives extradited by the US to India between 2002-2018, against 60 pending Indian requests in that period

- **India's Discretion on Timing**

- Even if treaty conditions are satisfied, extradition is not automatic; Section 31 of the Extradition Act allows India to postpone surrender while domestic proceedings are pending

- **The Process**

- A formal US request goes through the MEA, is examined against treaty/Act requirements, certified as extraditable by an Indian court, with the final surrender decision resting with the Union government

Bishnoi Indictment: The Long Road to Extradition

- **Key Provisions of the Act**
 - **Applicability:** Covers both treaty-based and non-treaty extradition cases.
 - **Dual Criminality Principle:** The offense must be a crime in both India and the requesting country.
 - **Extradition Offenses:** Covers terrorism, organized crime, drug trafficking, and financial fraud.
 - **Bar on Political Offenses:** Individuals cannot be extradited for political offenses.
 - **Rule of Specialty:** The fugitive can only be tried for the crime they were extradited for.
 - **Human Rights Considerations:** No extradition if there is a risk of torture or an unfair trial.
 - **Nodal Authority for Extradition**
 - **Ministry of External Affairs (MEA):** Processes extradition requests from foreign governments.
 - **Central Bureau of Investigation (CBI):** Handles extradition-related investigations.
 - **National Investigation Agency (NIA):** Pursues cases related to terrorism and national security.

GCC Is Office Success Story. Now, Focus on Factory

GCC is office success story. Now, focus on factory

FOREIGN DIRECT investment (FDI) in China has gone mainly into manufacturing — initially labour-intensive and then hi-tech. Not for nothing it has earned the moniker of “the world’s factory”. FDI in India has come, and continues to do so, largely in services, making it “the world’s office” for global corporations. According to Finance Minister Nirmala Sitharaman, India now hosts 2,100-plus global capability centres (GCC) of over 500 of the Forbes top-2,000 companies. These GCCs together employ some 23 lakh professionals and generate nearly \$100 billion in annual revenue. Moreover, they seem to have evolved beyond being cost-saving “back offices” handling basic IT, finance and customer services for multinational firms. Some of the GCCs have become strategic hubs for MNCs to build technology and enterprise platforms, undertake core engineering research and development, and work on clinical research and accelerated drug discovery using artificial intelligence.

The GCC success story — India is home to half of such centres worldwide and adding one every day — is also reflected in the country’s external balance of payments profile. In 2025-26, its exports of services, at \$421.3 billion, was close to the export of goods worth \$446.1 billion. On the other hand, imports of goods (\$783.4 billion) were way above the imports of services (\$204.7 billion). Thus, while India recorded a merchandise trade deficit of \$337.3 billion, it had a surplus of \$216.6 billion on the services account. That’s a clear demonstration of where the country’s comparative advantage lies — not in factories, but in offices. The GCCs and the traditional Indian IT services companies are in the driver’s seat.

There’s a flip side to this. The GCCs and Indian IT outsourcing are concentrated in Bengaluru, Delhi-NCR, Hyderabad, Chennai, Mumbai and Pune. Sitharaman wants them to come up more in Varanasi, Visakhapatnam, Mysuru, Tiruchirappalli and other tier-2 and tier-3 cities. That’s not easy. Like all industries, GCCs/IT firms tend to cluster around major metro areas that offer both robust infrastructure (international airports, dedicated office parks, reliable transport, power and utilities) and talent density (skilled engineers, developers and managers). Replicating this ecosystem in smaller centres may not be as feasible, unlike manufacturing clusters that can thrive in semi-urban and rural areas having lower land and labour costs. The constraint there is only physical infrastructure. India cannot afford to give up on manufacturing — whether textiles, leather, agro-processing, gems and jewellery — that alone can absorb the mass of its less-skilled workforce. India needs both factories and offices.

GCC Is Office Success Story. Now, Focus on Factory

- **India as "The World's Office"**

- Unlike China's FDI-driven manufacturing dominance, India's FDI has flowed largely into services; India now hosts 2,100+ Global Capability Centres (GCCs) of over 500 Forbes top-2000 companies

- **Scale and Evolution**

- These GCCs employ around 23 lakh professionals and generate nearly \$100 billion in annual revenue; many have evolved beyond cost-saving "back offices" into strategic hubs for R&D and AI-driven drug discovery

- **Trade Balance Evidence**

- In 2025-26, India's services exports (\$421.3 billion) nearly matched goods exports (\$446.1 billion); India ran a \$337.3 billion merchandise trade deficit but a \$216.6 billion services surplus

- **Geographic Concentration**

- **GCCs remain concentrated in Bengaluru, Delhi-NCR, Hyderabad, Chennai, Mumbai and Pune; the Finance Minister wants expansion into tier-2/3 cities, though replicating the ecosystem there is challenging**

- **The Manufacturing Imperative**

- India cannot give up on manufacturing (textiles, leather, agro-processing, gems and jewellery) since it alone can absorb the mass of India's less-skilled workforce :India needs both factories and offices

Five Crore Indians Wait When the Courts Take a Break

Five crore Indians wait when the courts take a break

The Supreme Court of India and the major High Courts are on a summer holiday break, i.e., shut down with some vacation Benches functioning to hear urgent matters. However, shops, markets, eating places, government and private offices, hospitals and police stations do not cease to work even though their staff enjoy personal leave. On the other hand, somewhere in an Indian jail tonight, a man who has been convicted of nothing is finishing another year behind bars, his trial still ongoing and perhaps years away from ending. He is one of the roughly three in four prisoners in this country who are undertrials: unconvicted, presumed innocent, yet serving time anyway – often longer than the sentence they would have received had they pleaded guilty. And as you read this, the courts, the public institutions that hold the fate of the poor in their hands, are functioning at only a fraction of their strength.

A three-century backlog

From June 1 to July 12, the Supreme Court is on its summer break, sitting with three or four Benches a week instead of a full court and resuming in earnest on July 13. The doors are not bolted; urgent pleas are still heard. But for six weeks, the most powerful court in the world's most populous democracy idles.

Now hold that against the number it is meant to be fighting. As of the last day of 2025, more than 5.39 crore cases were pending in Indian courts: over 4.76 crore in the district courts, 63.6 lakh in the High Courts, and more than 92,000 in the Supreme Court itself, its heaviest load in over 30 years. A government study once calculated that, at the present pace, clearing the pile could take three centuries.

Three centuries. And the Court takes the summer and winter vacations off.

To be fair, the judges deserve it. The recess is when reserved judgments finally get written, when the brutal daily cause-list eases enough to think



Vijay L. Kelkar

Vice-Chairman of the Pune International Centre



Pradeep S. Mehta

Secretary-General of CUTS International (Consumer Unity and Trust Society)

When courts rest together for vacations, crores of litigants across the country wait longer for justice

Last year the Chief Justice of India and his four senior-most colleagues worked clean through the first week of the break. Indian judges are among the most overworked in the world. This is not an accusation of laziness; no one who has seen a judge's docket could make one.

Because that is precisely the point. When the people doing the work are exhausted and the backlog is biblical, what conceivable logic shuts the whole shop at once? Not that judges rest – which they should – but that almost all of them rest together, so the institution goes quiet for six-plus weeks every year. And the habit is a colonial hand-me-down: the calendar was built for English judges who could not stand the Indian heat and retreated to cooler climes each summer, then enjoyed long Christmas holidays in the winter.

In 2024, the Court announced a fix: it renamed the “summer vacation” as “partial court working days”. That was it. The number of days it actually sits – about 190 days in a year – did not change. You cannot rebrand your way out of a backlog; a litigant whose case is stuck does not care what the recess is called, only whether his matter is heard and disposed of.

Justice needs court continuity

So what would actually help? The first answer is one the judiciary's own watchdogs have urged for years: do not abolish judicial rest; stagger it. A parliamentary standing committee in 2023 objected to “the entire court going on vacation en masse”, and proposed the obvious: rotate the leave, keep the Benches full, and never let the court run thin. A hospital does not empty its wards because its doctors are owed time off; it builds a roster. The Law Commission and the Justice Malimath Committee said the same, long ago. They were not enemies of the courts; they were trying to save them from themselves.

The standard objection is that all this is a sideshow: the real disease is vacancies, not vacations. It is true that courts are starved of judges, with up to a third of High Court seats

lying empty. But the objection boomerangs. If the Bench is already running at half strength, thinning it further for six weeks every summer is not a defence of the recess; it is the strongest case against it. And while filling those seats depends on the government and the collegium and will drag on for years, the calendar is the judiciary's to fix. One reform needs permission; the other needs only will.

Beyond courts, towards solutions

Staggering the leave, though, only treats the symptom. The deeper fix is to keep disputes out of the courtroom in the first place; it was never meant to be the first stop for every quarrel, but the last. India already has the tools. Lok Adalats settled more than 2.59 crore cases in a single national sitting last December, and over 23.5 crore in three years. The Mediation Act of 2023 now nudges parties to attempt settlement before going to court, and arbitration can take commercial disputes entirely out of judges' hands; yet these routes remain badly underused. But none of this happens on its own.

The country has a reservoir of experience it scarcely draws on: its retired judges, who spent their careers mastering this very machinery and step down, still in full command of it, at 62 or 65. Their expertise need not retire with them. Albeit many of them do function as heads of quasi-judicial bodies and tribunals. Freed from any daily docket, a dedicated corps of former judges could be invited to take up a single targeted charge: identify where cases pile up, set public targets for their disposal, and account for progress openly. Reform that belongs to everyone belongs to no one.

The judges will keep working through the heat, as they always do. The honest demand is not that they stop resting, but that the courts stop closing ranks and falling silent at the same moment, because justice, for five crore Indians, is not on vacation. It is overdue.

With contributions from Anushka Kewlani of CUTS

Five Crore Indians Wait When the Courts Take a Break

- **The Scale of Pendency**

- As of end-2025, over 5.39 crore cases were pending in Indian courts (4.76 crore in district courts, 63.6 lakh in High Courts, over 92,000 in the Supreme Court : its heaviest load in over 30 years)

- **The Vacation Practice**

- The Supreme Court and major High Courts take a six-week-plus summer break annually with only limited vacation Benches functioning; the practice traces to English judges retreating from Indian heat

- **The 2024 "Fix" That Wasn't**

- The Court renamed "summer vacation" as "partial court working days" in 2024, but the actual number of sitting days (about 190/year) remained unchanged, a rebranding, not a structural fix

- **Recommended Reform : Staggered Leave**

- A 2023 parliamentary standing committee, the Law Commission and the Justice Malimath Committee have all proposed rotating leave among judges rather than the entire court going on vacation en masse

China's Long March in Space Tech Continues With Reusable Rocket

China's Long March in space tech continues with reusable rocket

Arav Shah

New Delhi, July 12

CHINA ON Friday successfully landed a reusable rocket for the first time — a milestone for the country's space programme that will challenge American companies' dominance in the technology.

The Long March 10B rocket — which can carry a payload of up to 16 metric tons to low Earth orbit — lifted off from the Wenchang Commercial Space Launch Site in Hainan. Low Earth orbit is a region between 160 km and 2,000 km above the Earth.

Six minutes after liftoff, the rocket's booster — the part that houses the engine and provides the initial thrust — separated from its upper stage and began a controlled descent to a floating platform on the sea. There, it was caught by a net.

Before China, only Elon Musk's SpaceX and Jeff Bezos's Blue Origin demonstrated the ability to launch and land rockets.

The mechanics of re-entry

All rockets lift off in the same way. Multiple segments, beginning with the first stage, power its ascent one by one before being discarded and destroyed. A reusable rocket, however, is one that can land on Earth after lifting off, thus allowing its valu-

able booster to be used again. This substantially reduces the cost of a launch and, by extension, space exploration.

How does the Chinese rocket differ?

There is one key difference between the SpaceX-Blue Origin reusable rockets and China's Long March — the final capture mechanism. Traditional reusable rockets rely on hydraulic landing "legs" to touch down on a ground pad or floating platforms on the sea. Space X pulled off a major feat in 2024 when it used giant mechanical arms — termed "chopsticks" — to catch its Starship rocket booster out of the air.

The Long March, though, uses a different system. Its onboard computer coordinates with a maritime landing vessel, which, in turn, aligns a suspended wire net directly below the descending rocket. The rocket is equipped with "landing hooks" that catch the net as the engines shut down.

Why does this test matter?

Space exploration is expensive. Reusable rockets make this process economical. China went a step further by removing the hydraulic landing legs in favour of the landing hooks — reducing the rocket's mass and thus increasing its payload margin.

Anushka Saxena, Research Analyst at



China's Long March 10B rocket lands on a floating platform on Friday. AP

the Takshashila Institution's Geostrategy programme told *The Indian Express*: "The technological and commercial base in China is faring very well despite having been shaken up by internal wrangling and corruption in the past."

This test comes as the global aerospace sector races to deploy massive commercial satellite constellations and prepares for crewed lunar landings, and puts China in direct competition with SpaceX.

Here's a look at the various strategic implications of the Chinese milestone:

LUNAR RACE: The Long March 10 series

of rockets is central to China's plans to put astronauts on the moon by 2030. This timeline directly competes with the American Artemis program, escalating the stakes of the race to secure real estate at the Moon's South Pole. This is where permanently shadowed craters contain billions of tonnes of ice water that can be used to create rocket propellant — making them key for any future deep space exploration.

LOW-EARTH ORBIT MISSIONS: Low Earth orbit missions are important for Earth observation and communication. China plans to deploy three communications satellite constellations comprising thousands of satellites each. Deploying such large numbers through traditional rockets is economically inviable. The Long March 10B's success (and its 16-tonne payload) provides the cost-effectiveness required to challenge Western rivals such as SpaceX.

"SpaceX and the Chinese are competing on two fronts — the frequency of launches and the capacity to fill the lower-earth orbit. The Chinese have demonstrated that they can do whatever the Americans can and are actively bridging the gap that SpaceX had created a decade ago," Saxena said, adding, "For other space-faring nations like India, it might take a long time before they can conclusively enter this circle."

China's Long March in Space Tech Continues With Reusable Rocket

- **The Milestone**

- China successfully landed a reusable rocket for the first time; the Long March 10B, capable of carrying up to 16 metric tons to Low Earth Orbit, lifted off from the Wenchang Commercial Space Launch Site in Hainan

- **Why Reusability Matters**

- Reusable rockets substantially cut launch costs; before China, only SpaceX and Blue Origin had demonstrated the ability to launch and land rockets

- **Strategic Implications**

- The Long March 10 series is central to China's plan to put astronauts on the Moon by 2030, directly competing with the American Artemis programme for access to the Moon's water-ice-rich South Pole

- **Commercial Competition**

- China plans large low-Earth-orbit satellite constellations; the Long March 10B's cost-effectiveness and 16-tonne payload position it to challenge Western rivals like SpaceX in launch frequency and orbital capacity

China's Long March in Space Tech Continues With Reusable Rocket

- **Reusable Launch Vehicle – Technology Demonstrator (RLV-TD) (Pushpak)**: A winged technology demonstrator that simulates an aircraft-like landing. Indian Space Research Organisation (ISRO) successfully conducted the LEX (Landing Experiment) series (LEX-01, 02, 03) using the 'Pushpak' vehicle to test autonomous horizontal landing.

Thank You